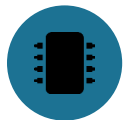


EEUC201 | ANALOG & DIGITAL ELECTRONICS

Operational Amplifiers

The 741 IC — Fundamentals, Specifications & Feedback

Introduction to ICs • Pin & Block Diagrams • Ideal Op-Amp • Datasheet Parameters • Feedback • Closed-Loop Gain



Introduction to Integrated Circuits

An Integrated Circuit (IC) is a miniaturized electronic circuit — transistors, resistors, capacitors and diodes — fabricated together on a single chip of semiconductor material (usually silicon). ICs reduce size, cost, and power consumption while improving reliability compared to discrete circuits.

Classification of ICs

- By fabrication: Monolithic ICs vs Hybrid ICs
- By function: Analog (Linear), Digital, and Mixed-Signal ICs
- By scale of integration: SSI, MSI, LSI, VLSI, ULSI
- By application: General-purpose (e.g., 741 op-amp) vs Special-purpose ICs



Analog IC

Continuous signals
(e.g., 741 Op-Amp)



Digital IC

Binary logic levels
(e.g., logic gates)



Mixed-Signal

Combines analog
+ digital blocks



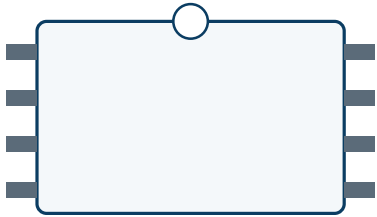
Monolithic

Entire circuit on
one silicon chip

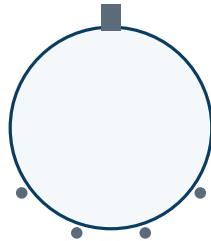


IC Packages & the Op-Amp Symbol

Common IC Packages



DIP-8 (Dual-in-line)

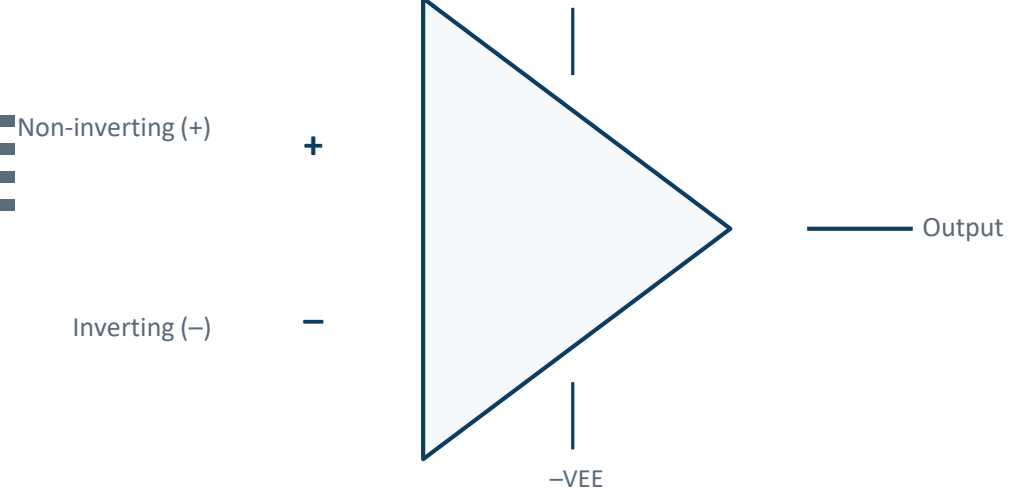


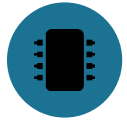
TO-99 (Metal Can)



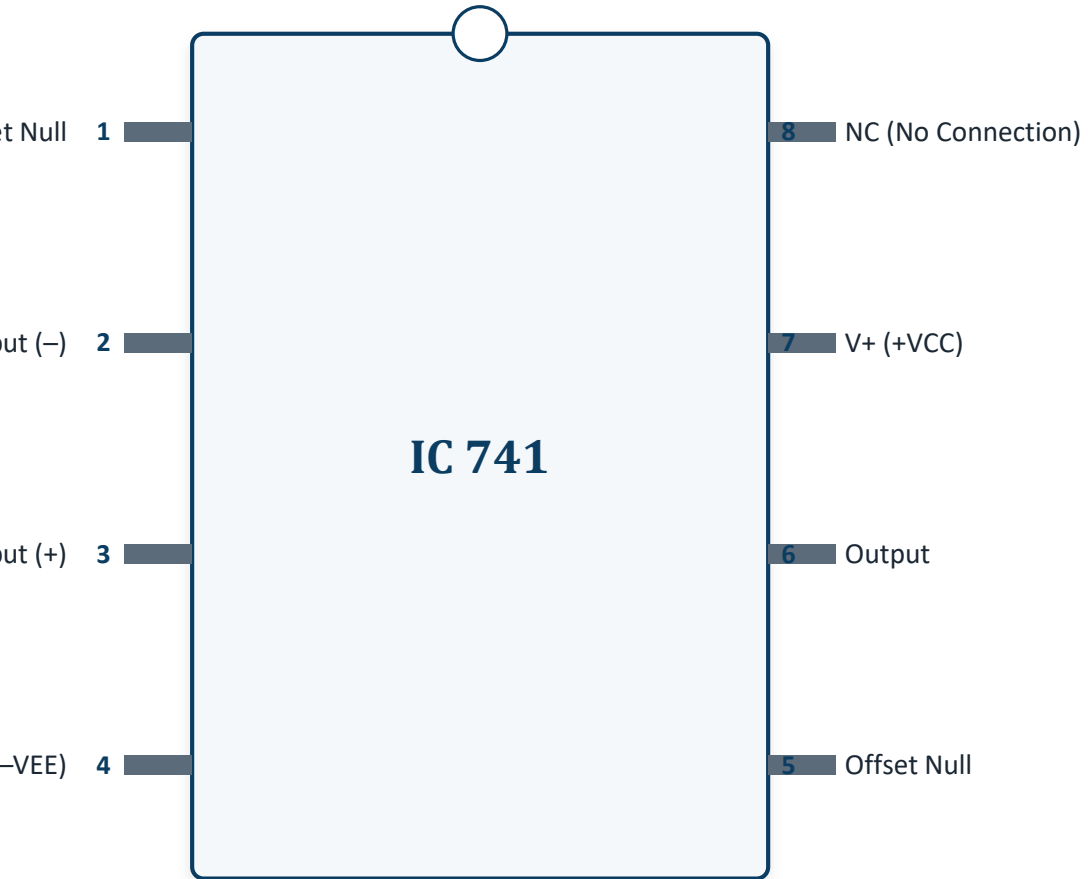
SOIC (SMD)

Standard Op-Amp Symbol





Pin Diagram of the 741 Op-Amp (8-Pin DIP)

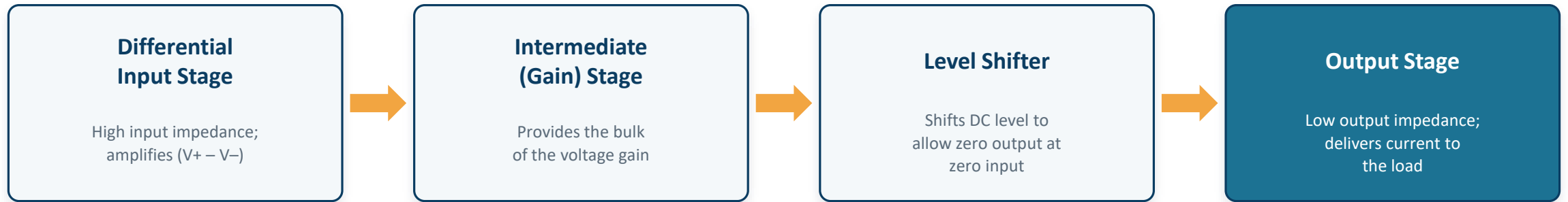


Key Points

- Pins 2 and 3 are the two signal inputs; pin 6 delivers the amplified output.
- Pins 7 and 4 connect the dual (split) power supply, typically ± 15 V.
- Pins 1 and 5 allow external offset-null trimming using a potentiometer.
- Pin 8 is unused (No Connection) in the standard 8-pin DIP package.



Internal Block Diagram of a Typical Op-Amp



Signal flow: the differential input stage senses the difference between the two inputs; the intermediate stage provides most of the open-loop voltage gain; the level shifter re-centers the DC level; the output stage supplies a low-impedance drive to the external load.

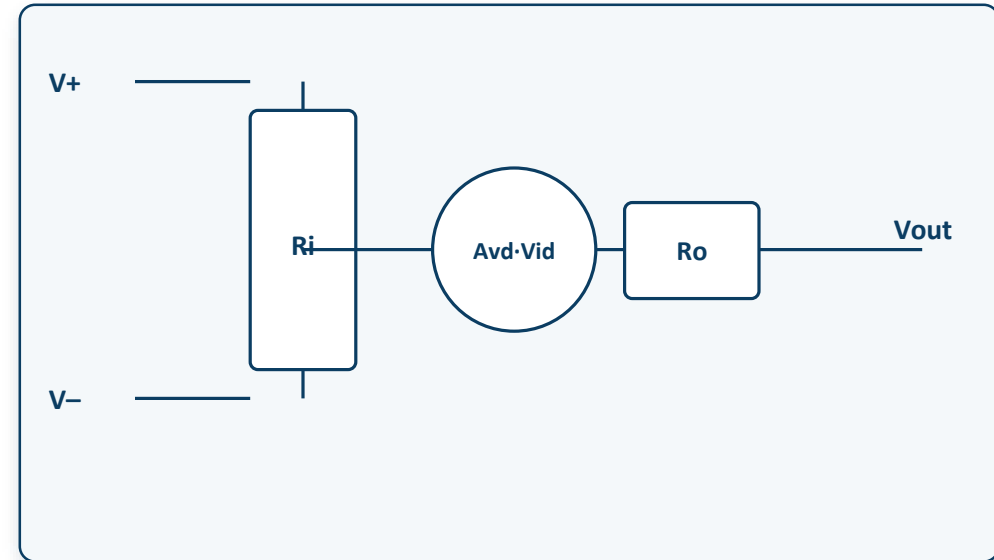


The Ideal Op-Amp & Its Equivalent Circuit

Ideal Characteristics

- Infinite open-loop voltage gain ($A_{vd} \rightarrow \infty$)
- Infinite input resistance ($R_i \rightarrow \infty$), zero input current
- Zero output resistance ($R_o \rightarrow 0$)
- Infinite bandwidth (gain constant at all frequencies)
- Infinite CMRR (rejects common-mode signals fully)
- Zero offset voltage/current and no drift with temperature

Equivalent Circuit





Specifications & Datasheet Parameters (1/2)

Parameter	Symbol	Typical (741)	Significance
Differential Input Resistance	R_i	$\approx 2 \text{ M}\Omega$	Impedance seen between the two input terminals
Output Resistance	R_o	$\approx 75 \Omega$	Internal impedance looking into the output terminal
Common Mode Rejection Ratio	CMRR	$\approx 90 \text{ dB}$	Ability to reject signals common to both inputs
Slew Rate	SR	$0.5 \text{ V}/\mu\text{s}$	Maximum rate of change of output voltage
Power Supply Rejection Ratio	PSRR	$\approx 96 \text{ dB}$	Immunity of output to supply-voltage variations

R_i and R_o describe the op-amp's input/output impedance behaviour; CMRR and PSRR quantify its noise immunity to common-mode and supply disturbances; Slew Rate limits how fast the output can respond to large, fast input changes.



Specifications & Datasheet Parameters (2/2)

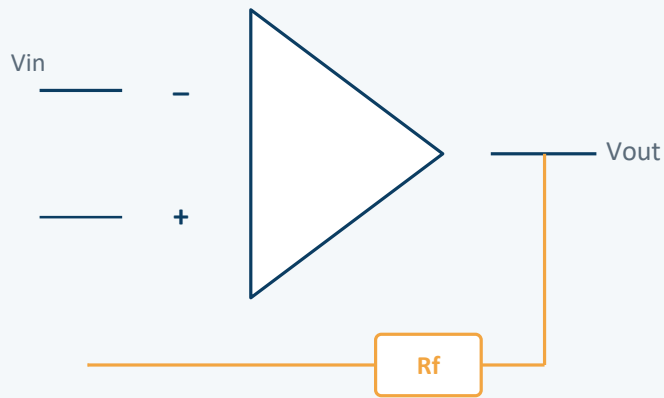
Parameter	Symbol	Typical (741)	Significance
Input Offset Voltage	V_{io}	$\approx 2 \text{ mV}$	Differential DC voltage needed for zero output
Input Offset Current	I_{io}	$\approx 20 \text{ nA}$	Difference between the two input bias currents
Input Bias Current	I_{ib}	$\approx 80 \text{ nA}$	Average DC current flowing into the two inputs
Output Offset Voltage	V_{oo}	$V_{io} \times (1 + A_{CL})$	Resulting DC error appearing at the output

These four “offset” parameters arise from unavoidable mismatches inside the differential input stage. They cause a small, unwanted DC error at the output even when the input is truly zero — offset-null pins (1 and 5) allow this error to be trimmed out externally.

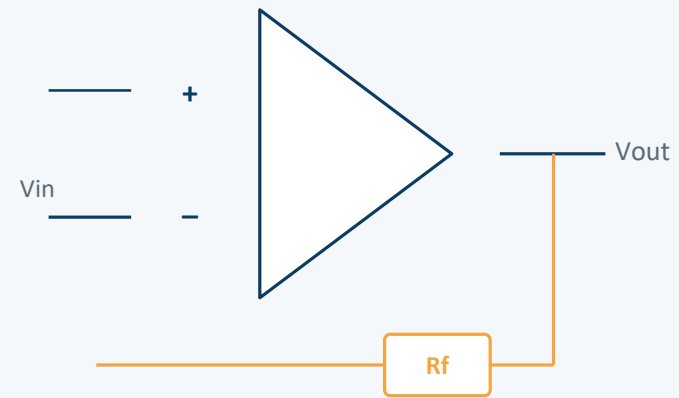


Feedback Configurations

Inverting Amplifier



Non-Inverting Amplifier



In both configurations, a fraction of the output is fed back to the inverting input through R_f . Negative feedback trades excess open-loop gain for stability, wider bandwidth, lower distortion, and a precisely defined closed-loop gain.



Op-Amp With and Without Feedback

Characteristic	Without Feedback (Open-Loop)	With Negative Feedback (Closed-Loop)
Voltage Gain	Very high ($\approx 2 \times 10^5$), but poorly defined	Precisely set by external resistors (R_f , R_{in})
Bandwidth	Narrow	Widened (gain-bandwidth product is constant)
Stability	Sensitive to temperature, ageing, supply drift	Highly stable and predictable
Distortion / Noise	Higher non-linearity	Significantly reduced
Input / Output Impedance	Very high R_i , non-zero R_o	R_i further increased, R_o further reduced

= Closed-Loop Voltage Gain

Non-Inverting Amplifier

$$ACL = 1 + R_f / R_{in}$$

Output is in phase with the input; gain is always ≥ 1 .

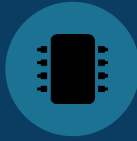
Inverting Amplifier

$$ACL = - R_f / R_{in}$$

Output is 180° out of phase with the input.

Why the Formula Holds (Ideal Op-Amp Assumptions)

- Virtual short: with negative feedback, $V_+ \approx V_-$ at the input terminals
- Virtual open: no current enters either input terminal ($R_i \rightarrow \infty$)
- Applying KCL at the inverting node and solving for V_{out} / V_{in} gives the closed-loop gain above



Key Takeaways

- The 741 is a general-purpose monolithic op-amp housed in an 8-pin DIP package.
- Its internal structure follows a standard flow: differential input → gain stage → level shifter → output.
- Ideal assumptions (infinite R_i and gain, zero R_o) simplify analysis; real datasheet parameters (CMRR, Slew Rate, offsets) capture non-ideal behaviour.
- Negative feedback trades raw open-loop gain for stability, bandwidth, and a precisely defined closed-loop gain: $ACL = 1 + R_f/R_{in}$ (non-inverting) or $-R_f/R_{in}$ (inverting).